Record 1 — Automated IoT Environmental & Cold-Chain Telemetry Log

System Log ID	**SYS-LOG-ZB-202605/06**
Storage Chamber	Cold Storage Zone B — Rack Bay R-04-12
Target Storage Profile	**Profile B — Chilled Saturated Saline** Operating envelope: +0.0°C to +4.0°C Strict band: +1.0°C to +3.5°C
Monitoring Duration	01 May 2026 — Inbound Put-Away to 05 June 2026 — Pick-Release
IoT Telemetry Gateway	SCADA-Chiller-Node-ZB4
Sensors / Calibration	Dual calibrated PT100 thermistors. NIST Calibration Reference: **CAL-TEMP-2026-0418**

SCADA / IoT CONTINUOUS COLD-CHAIN SURVEILLANCE

Monitored Node	Zone B / R-04 Tier 2 — Sensors SN-T04-A and SN-T04-B
Sampling Frequency	Every 5 minutes — real-time cloud push to WMS
Total Logged Dwell	**840 hours — 35 calendar days**
HACCP Action Limit	**Instantaneous > +6.0°C OR > +4.0°C for > 15 minutes**

1. Cumulative Dwell Period Telemetry Summary

Reporting period: 01 May–04 June 2026

Parameter	Engineered Specification	Minimum	Maximum	Mean	Sensor Status / Finding
Zone B Core Ambient Temperature	+1.0°C to +3.5°C	+1.2°C	+2.4°C	+1.82°C	Online / Dual Sync **CONFORMING — PASS**
Relative Humidity	85% to 90%	86.1%	89.4%	87.8%	Online **CONFORMING — PASS**
Defrost Cycle Spikes	≤ +4.0°C for < 15 minutes	+1.8°C	+3.4°C	+2.7°C Maximum duration: 9 minutes	Intelligent Pulse **CONFORMING — PASS**
Refrigerant Line Pressure	14.2 to 16.0 bar	14.6 bar	15.3 bar	14.9 bar	Normal **CONFORMING — PASS**
F-Gas Halogen Leak Sensor	< 50 ppm	0 ppm	0 ppm	0 ppm	Normal — Zero Leak **CONFORMING — PASS**

2. Micro-Interval Telemetry Extract — Order Pick Day

05 June 2026 | Sales Order LL628-0626 | All times HKT

The table presents 15-minute sample intervals leading up to and during pick-wave release. The stated underlying system sampling frequency is 5 minutes.

Time	Node A °C	Node B °C	RH %	Roll-Up Door	Compressor	Action / Alarm
06:00	+1.7	+1.8	88.1	Closed / Sealed	Primary Lead — Normal	None — Normal
06:15	+1.8	+1.8	87.9	Closed / Sealed	Primary Lead — Normal	None — Normal
06:30	+1.9	+1.9	88.3	Closed / Sealed	Primary Lead — Normal	None — Normal

Time	Node A °C	Node B °C	RH %	Roll-Up Door	Compressor	Action / Alarm
06:45	+1.8	+1.7	87.5	Closed / Sealed	Primary Lead — Normal	None — Normal
07:00	+1.7	+1.7	88.0	Closed / Sealed	Primary Lead — Normal	None — Normal
07:15	+1.8	+1.9	88.4	Momentary Open — 22 seconds	Primary Lead — Normal	None — Air Curtain Active
07:30	+2.1	+2.2	86.9	Momentary Open — 34 seconds	Primary Lead — Normal	None — Pick-Wave Pull
07:45	+1.9	+1.8	87.6	Closed / Sealed	Primary Lead — Normal	None — Stabilized
08:00	+1.8	+1.8	88.2	Closed / Sealed	Primary Lead — Normal	None — Normal

System Diagnostic Assessment

Thermal Excursions	Zero excursions outside the +0.0°C to +4.0°C threshold during the entire 35-day holding lifecycle of Lot Jap682/1234.
Door Cycling	Remained strictly within the ≤ 45-second envelope.
Compressor Redundancy	Secondary compressor remained on hot standby with zero mechanical faults logged.

RECORDED SYSTEM AUTO-VALIDATION
STATUS: 100% PASS — UNBROKEN COLD CHAIN VERIFIED

Electronic Signature / Timestamp	WMS-SCADA-GATEWAY-AUTH-STAMP#69921_05JUN2026_08:01:14